

Wave optics

$$\rightarrow \Delta\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} (\mu \Delta x) \text{ if med other than air, } \mu R \cdot I$$

$$\rightarrow \vec{y}_{res} = \vec{y}_1 + \vec{y}_2 + \dots + \vec{y}_n, \quad y_{res} = A \cos(\omega t + \phi) = R \cos(\omega t + \phi)$$

$$\rightarrow \text{for coherent source} \quad A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi} \quad I \propto A^2 \propto (\text{slit width})^2$$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi, \quad \text{for incoherent source} \quad I = I_1 + I_2 + \dots + I_n$$

$\rightarrow$ When $\Delta\phi = 0^\circ$ or $360^\circ \rightarrow$ constructive interference

$$A_{max} = A_1 + A_2, \quad I_{max} = [\sqrt{I_1} + \sqrt{I_2}]^2$$

$\rightarrow$ When $\Delta\phi = 180^\circ \rightarrow$ destructive interference

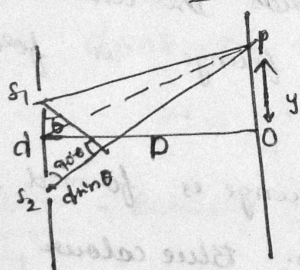
$$A_{min} = A_1 - A_2, \quad I_{min} = [\sqrt{I_1} - \sqrt{I_2}]^2$$

$$\frac{A_{max}}{A_{min}} = \frac{A_1 + A_2}{A_1 - A_2} = \frac{I_{max}}{I_{min}} = \frac{[\sqrt{I_1} + \sqrt{I_2}]^2}{[\sqrt{I_1} - \sqrt{I_2}]^2} \quad \frac{A_{max}^2}{A_{min}^2} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2}$$

$$\rightarrow I_{avg} = I_1 + I_2 + 2\sqrt{I_1 I_2} \langle \cos \phi \rangle \quad \text{but for long time } \langle \cos \phi \rangle = 0$$

$$I_{avg} = \frac{I_{max} + I_{min}}{2}$$

$\rightarrow$ YDSE



$$\Delta x = d \sin \theta \approx d \tan \theta = \frac{y d}{D} = \sqrt{D^2 + d^2} - D$$

$$\Delta x = (\mu_1 - 1) t$$

$$\Delta\phi = \frac{2\pi}{\lambda} (\Delta x)$$

$$R_{amp} = 2A \cos(\phi/2) \quad I_{res} = 4I_0 \cos^2(\phi/2)$$

$$\rightarrow \text{Bright fringe (max)}, \quad \Delta\phi = 2n\pi, \quad \Delta x = n\lambda, \quad I_{max} = 4I_0, \quad y_n = \frac{n\lambda D}{d}$$

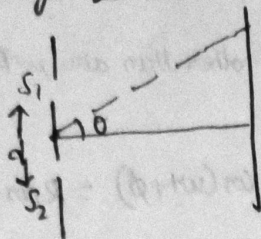
$$\rightarrow \text{Dark fringe (min)}, \quad \Delta\phi = \frac{(2n-1)\pi}{2}, \quad \Delta x = \frac{(2n-1)\lambda}{2}, \quad I_{min} = 0$$

$$y_n = \frac{(2n-1)\lambda D}{2d}$$

$$\rightarrow \text{Band width or fringe width} \quad \beta = \frac{\lambda D}{d}$$

$$\text{if YDSE conducted in other medium} \quad \beta_{med} = \frac{\beta_{free}}{\mu}$$

Angular fringe width (θ)



$$\theta = \frac{y}{D} = \frac{\lambda}{d} = \frac{\beta}{D}$$

$$\theta_{med} = \frac{\theta_{free}}{d}$$

b = size of object
 l = distance of object
 λ = wavelength of light

→ If $\frac{b^2}{\lambda} \gg 1$ then geometrical optics is satisfied.

Validity of Ray optics $Z_f = \frac{a^2}{\lambda}$
 → If $\frac{b^2}{\lambda} < 1$ or $\frac{b^2}{\lambda} \approx 1$ then wave optics is satisfied.

→ If n_1 fringes are observed with light of wavelength λ_1 & n_2 fringes are observed with light of wavelength λ_2 (bright bands)

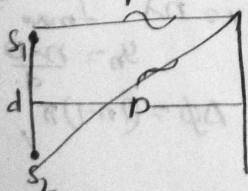
$$n_1 \lambda_1 = n_2 \lambda_2 \quad n_1 \beta_1 = n_2 \beta_2$$

$$\min. \quad n_1 \lambda_1 = (2n_2 - 1) \lambda_2$$

→ If white light ray is used in YDSE then all colours travel equally towards the centre of screen & hence white coloured fringe is formed at centre of screen.

→ $\lambda_R > \lambda_B$. Red colour fringe is formed far away from centre of screen. Blue colour fringe is visible nearer to centre of screen.

→ Screen placed $\parallel$ to plane of slits



$$d \sin \theta = n \lambda$$

If point sized sources are used hyperbolic fringe is formed.



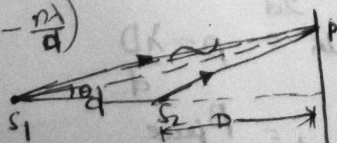
If cylindrical size

of lines.

If lines

cylindrical

$$= D \sqrt{2(1 - \frac{n \lambda}{d})}$$



$$d \cos \theta = n \lambda$$



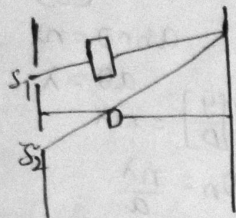
point size - concentric

→ Visibility (V) = $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$ $0 \leq V \leq 1$

$I_{\max} = I_{\min} \Rightarrow V = 0$ we can't observe dark & bright fringes clearly.

$I_{\min} = 0, \Rightarrow V = 1$ we can observe dark & bright fringe clearly.

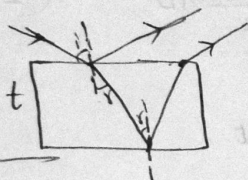
→ Lateral displacement of fringes:



$$y = \frac{(u-1)tD}{d} = \frac{(u-1)t\beta}{\lambda}, \beta = \frac{(u-1)tD}{d}$$

$$\text{no of fringes shifted } n = \frac{(u-1)t}{\lambda}$$

→ Inference in thin films:



$$2ut \cos r = \cancel{[n+1/2]\lambda}^{\text{max}} = [2n-1/2]\lambda^{\text{max}}$$

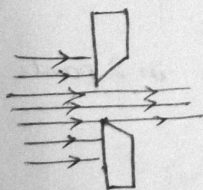
$$2ut \cos r = n\lambda_{\min} \Rightarrow n = 0, 1/2, \dots$$

→ Finding no of maxima in YDSE

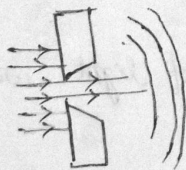
$$d \sin \theta = n\lambda \Rightarrow \sin \theta = \frac{n\lambda}{d}$$

$$-1 \leq \frac{n\lambda}{d} \leq 1 \Rightarrow -\frac{d}{\lambda} \leq n \leq \frac{d}{\lambda} \quad \text{minima is formed b/w 2 maxima.}$$

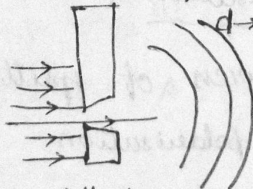
II. Diffraction - bending of light at sharp edges.



No diffraction
 $\lambda \ll d$

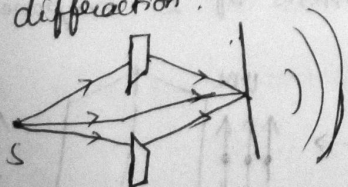


diffraction is noticeable
 $\lambda \sim d$



diffraction is noticeable
 $\lambda > d$

→ Fresnel diffraction:

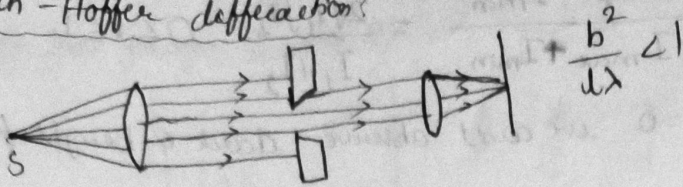


$$\frac{b^2}{l\lambda} \approx 1$$

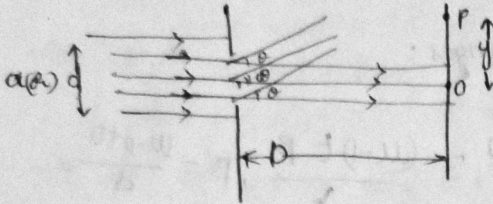
source & screen are very close

Mathematical approach is difficult.

Young-Hopper diffraction



Single slit diffraction



$$\Delta x = a \sin \theta = a \tan \theta = a \left[\frac{y}{D} \right]$$

$$a \sin \theta = n \lambda$$

$$a \theta = n \lambda$$

$$a \sin \theta = a \tan \theta = a \left[\frac{y}{D} \right] = n \lambda$$

for n^{th} order minima $y_n = \frac{n \lambda D}{a}$ $\theta_n = \frac{n \lambda}{a}$

for n^{th} order maxima $a \sin \theta = \left[\frac{2n+1}{2} \right] \lambda$

$$y_n = \frac{(2n+1)}{2a} \lambda D \quad \theta_n = \left(\frac{2n+1}{2} \right) \frac{\lambda}{a}$$

Resolving power = $\frac{1}{\text{Resolving limit}}$

Human eye $\theta = 1' \Rightarrow \text{limit}$

Telescope $\theta = \frac{1.22 \lambda}{d}$

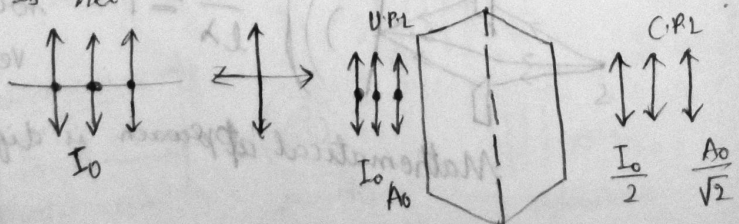
Microskop $\theta = \frac{0.61 \lambda}{\mu \sin \alpha}$

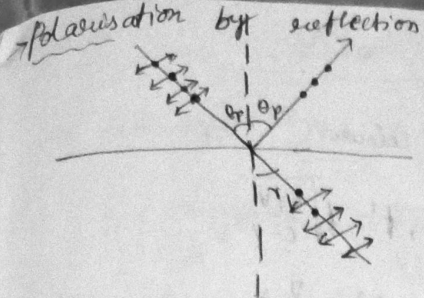
III. Polarisation

→ The process of splitting of light wave in 2 parts is known as polarisation.

→ In electromagnetic wave the components of electrical vectors are mutually perpendicular.

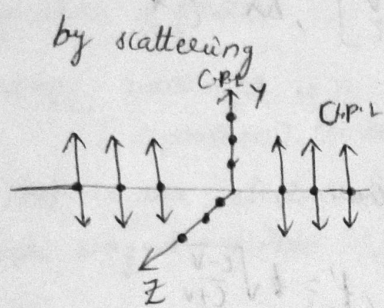
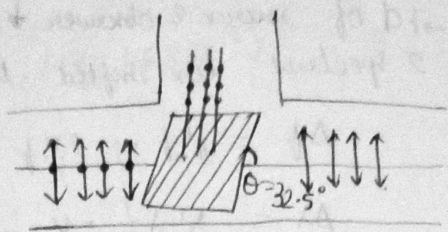
→ Unpolarised light (UPL) consists of 2 components 1 is parallel & other is perpendicular.





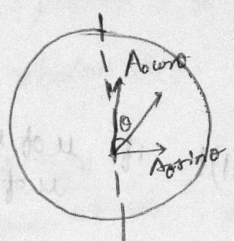
$$\tan \theta_p = \mu_2$$

by reflection



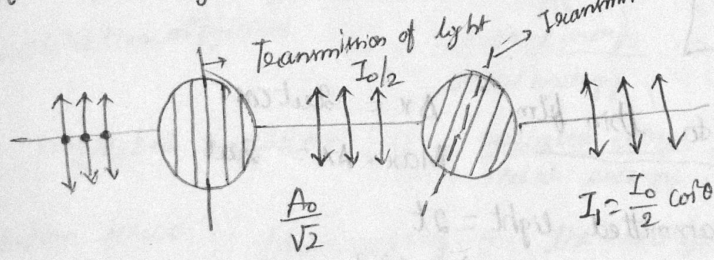
degree of polarisation depends on the angle of scattering

→ Effect of polariser on polarised light: only 1 polariser



$$I = \frac{I_0}{2} \quad A = \frac{A_0}{\sqrt{2}}$$

→ Effect of analyser on polarised light



$$I_1 = \frac{I_0}{2} \cos^2 \theta_1 \quad A = \frac{A_0}{\sqrt{2}} \cos \theta_1$$

$$I_2 = \frac{I_0}{2} \cos^2 \theta_1 \cos^2 \theta_2 \quad A = \frac{A_0}{\sqrt{2}} \cos \theta_1 \cos \theta_2$$

$$I = \frac{I_0}{2} \cos^2 (m-1) \theta$$

Doppler effect

1. d change ν, λ also changes.
2. Doppler effect in sound is asymmetric. In light is symmetric.
3. Only blue & red shifts are possible.

Brewster's Law



→ Blue shift

→ d of source & observer $\downarrow \Rightarrow \lambda \downarrow$

2. Spectral line shifted to blue & violet colour.

$$\Delta f = f' - f = +\frac{v}{c} f \Rightarrow f' = f \left[1 + \frac{v}{c} \right], f' = f \sqrt{\frac{c+v}{c-v}}$$

$$\Delta \lambda = \lambda' - \lambda = -\frac{v}{c} \lambda \Rightarrow \lambda' = \lambda \left[1 - \frac{v}{c} \right], \Delta \lambda = -\frac{v}{c} \lambda$$

→ Red shift

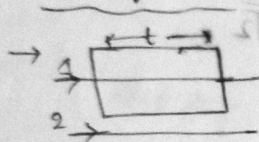
1. d of source & observer $\uparrow \Rightarrow \lambda \uparrow$

2. Spectral line shifted to red colour.

$$\textcircled{1} \Delta f = f' - f = -\frac{v}{c} f \Rightarrow f' = f \left[1 - \frac{v}{c} \right], f' = f \sqrt{\frac{c-v}{c+v}}$$

$$\textcircled{2} \Delta \lambda = \lambda' - \lambda = +\frac{v}{c} \lambda \Rightarrow \lambda' = \lambda \left[1 + \frac{v}{c} \right]$$

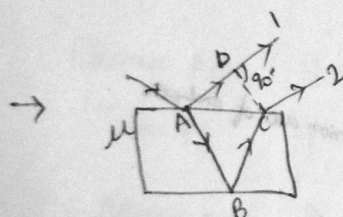
→ Interference



t = thickness

$$\Delta x = (\mu - 1)t$$

$$\Delta x = \left(\frac{\mu_2}{\mu_1} - 1 \right)t \quad \text{if } \begin{matrix} \mu \text{ of slab} = \mu_2 \\ \mu \text{ of med.} = \mu_1 \end{matrix}$$



$$\Delta x = \mu(ABC) - AD$$

$$t = n\lambda$$

⇒ YDSE due to thin film

$$\Delta x = 2\mu t \cos r$$

$$\text{Max. } \Delta x = 2\mu t$$

① Due to transmitted light = $2t$

i) Bright fringe $2\mu t \cos r = n\lambda, n=1,2,3,\dots$

ii) dark fringe $2\mu t \cos r = (2n-1)\frac{\lambda}{2}, n=1,2,3,\dots$

② Due to reflection

i) Bright fringe $2\mu t \cos r + \frac{\lambda}{2} = n\lambda, n=1,2,3,\dots$

$$2\mu t \cos r = (2n-1)\frac{\lambda}{2}, n=1,2,3,\dots$$

ii) dark fringe $2\mu t \cos r + \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2}, n=0,1,2,3,\dots$

$$2\mu t \cos r = n\lambda, n=1,2,3,\dots$$

→ for scattering size of obstacle should be smaller than wavelength, and here